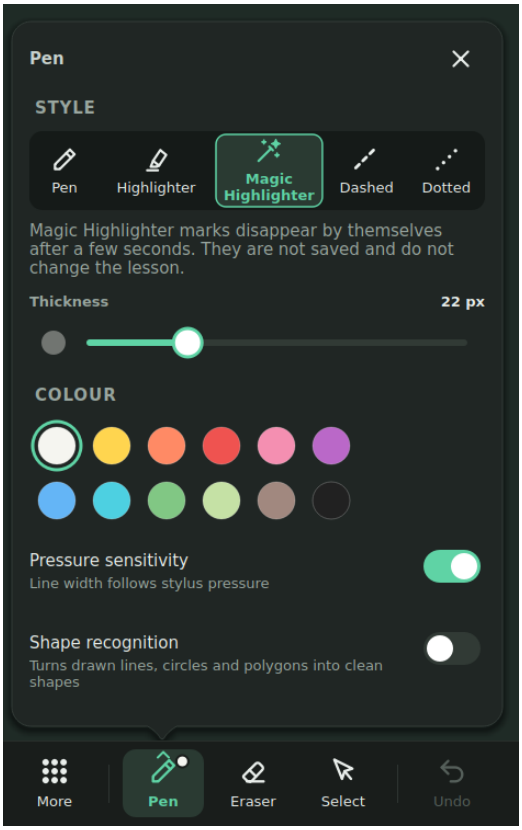
ClassBoard

How to use the app

The digital whiteboard for classroom touch boards: writing, erasing, geometry, equations, graphs, calculator, pages and lessons, explained step by step.



The Pen menu, opened from the ribbon at the bottom of the board.

Version 0.3.0 · Windows 10 / 11 · works fully offline

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Good to know: Everything in ClassBoard works without the internet. Nothing you write or import is uploaded anywhere.

1. Installing and starting ClassBoard

Install

1. Download **ClassBoard-Setup-<version>-x64.exe** (from the project's GitHub Releases page, or the ClassBoard-Setup-windows-x64 build download, which comes as a zip with the installer inside).
2. Double-click the installer. If Windows shows **“Windows protected your PC”**, click **More info → Run anyway**. This appears because the installer is not digitally signed.
3. Choose **Install for all users** (needs an administrator password; best for a classroom board) or **Only for me**.
4. Keep **Create a desktop shortcut** and **Open .classboard lessons with ClassBoard** ticked.
5. Click **Install**, then **Finish** to start ClassBoard.

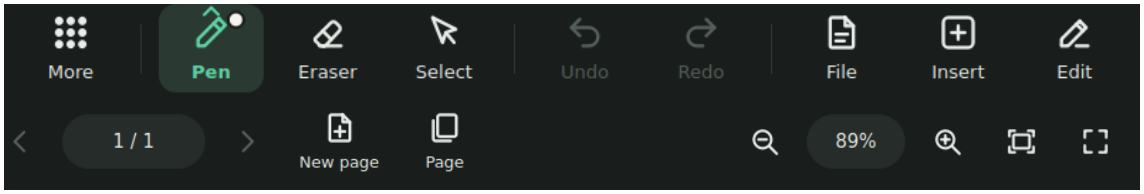
Start, update and remove

- Start ClassBoard from the **Start menu** or the **desktop shortcut**, or double-click a saved **.classboard** lesson.
- To **update**, run the newer installer. It replaces the old version and closes ClassBoard first if it is open. Your lessons and settings are kept.
- To **remove** it: Windows **Settings → Apps → ClassBoard → Uninstall**. Saved lessons (in your Documents folder), settings and autosave data stay.

> **Tip:** On a school computer, ask the IT team before installing. Installing for all users needs an administrator.

2. The screen and the ribbon

ClassBoard opens on a clean board with the pen ready. Everything is in the **ribbon** along the bottom. Buttons open small menus (popovers) right above them, so you never leave the board. Tap anywhere outside a menu to close it.



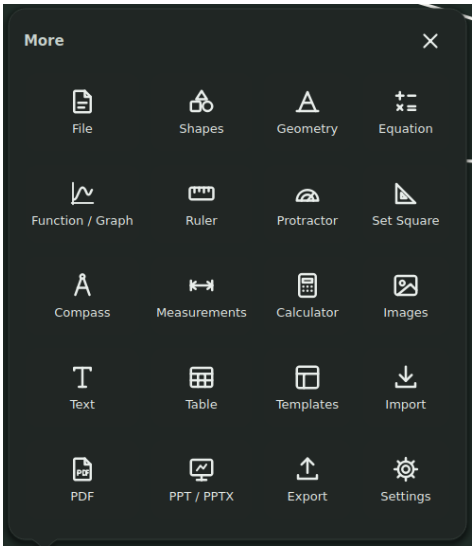
The ribbon at the bottom of the board. Top: More, Pen, Eraser, Select, Undo, Redo, File, Insert, Edit. Bottom: page arrows and page number, New page, Page, zoom out, zoom %, zoom in, fit page, full screen.

Button	What it does
More	Every other tool: shapes, geometry, equations, graphs, instruments, images, text, tables, templates, import, export, settings. A tool you pick from More appears as an extra ribbon button.
Pen / Eraser / Select	Switch tools. Tap the **active** tool again to open its options (a small dot on the button shows options are available).
Undo / Redo	One history for everything you do. Undo also jumps to the page it changes.
File	New, open, save, import, export.
Insert	Image, shape, text, graph, equation, table, PDF, PowerPoint.
Edit	Undo, redo, cut, copy, paste, duplicate, delete, select all.
‹ 3 / 12 ›	Previous / next page. Tap the number to open the page navigator.
New page / Page	Add a blank page / open the page menu.
− 81 % +	Zoom out, the View menu, zoom in. The frame button fits the page; the last button is full screen.

The More menu holds all tools in one place:

- File, Shapes, Geometry, Equation
- Function / Graph, Ruler, Protractor, Set square
- Compass, Measurements, Calculator, Images
- Text, Table, Templates, Import
- PDF, PPT / PPTX, Export, Settings

On a narrow screen File, Insert and Edit are hidden from the ribbon; everything is still in More.



The More menu

3. Touch, pen and mouse

Action	Touch	Pen	Mouse
Write / use the tool	One finger. Every finger on its own writes its own line (up to 10 or more at once).	Pen tip	Left button
Move the board (pan)	Two fingers of one hand, or one finger of each hand, dragged	–	Right or middle drag, Shift + wheel
Zoom	Pinch: two fingers of one hand, or one finger of each hand	–	Mouse wheel
Erase	Three fingers held together, or the flat of your hand, and wipe	Eraser end of the pen	–

Several people writing at once (ten-point touch)

Every finger that touches the board by itself writes its own line, so several students can write at the same time. A line that is really being drawn is never interrupted by another finger.

Zoom and move the board

- Put **two fingers of one hand** down close together (closer than about 16 cm) and pinch or drag. You can also use **one finger of each hand**, put down at the same moment (up to about 45 cm apart).
- Your fingers don't have to land at exactly the same time: the second finger can come up to about a third of a second later. The little start of a line from the first finger is removed automatically.
- Zooming does not start while somebody else is really writing (their line would bend). A finger simply resting on the board does not block it.
- Fingers that touch somewhere else while you zoom wait until you have finished.

Wipe like a board eraser

Put **three fingers side by side** (touching or nearly touching) and wipe. It works in every tool. It rubs out ink and parts of shapes, and removes equations and other objects it passes over; pictures and imported slides stay. It never draws, and the whole wipe is one Undo step. Fingers that are spread apart are never a wipe. The flat of your hand works too.

Writing with a pen

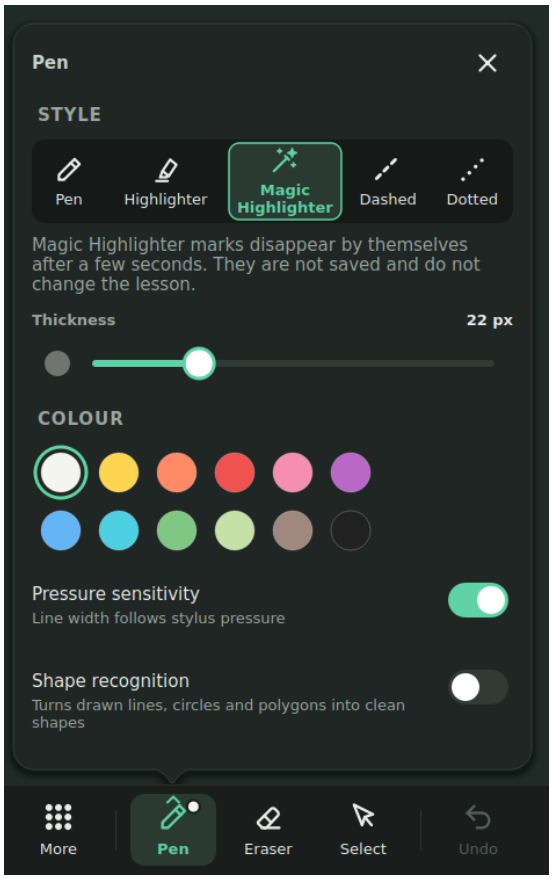
While the pen touches the board or hovers over it, touches are ignored, so the hand resting on the board neither draws nor erases. The eraser end of a pen erases.

Tip: If young pupils keep zooming by accident, turn on **Settings → Every finger draws**. Two-finger zoom is then off and every finger writes.

4. Writing: pen, highlighter and Magic Highlighter

Tap **Pen** to write. Tap it again to open the Pen menu:

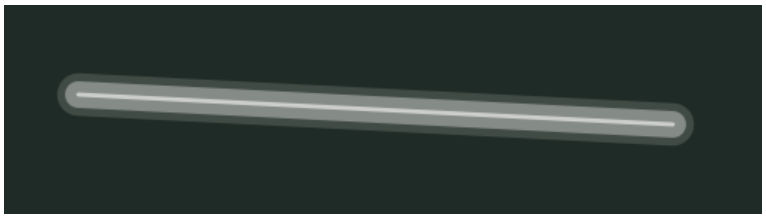
- **Style:** Pen, Highlighter, Magic Highlighter, Dashed, Dotted.
- **Thickness:** drag the slider; the dot shows the size.
- **Colour:** tap a colour.
- **Pressure sensitivity:** the line gets thicker when you press harder with a stylus.
- **Shape recognition:** roughly drawn lines, circles, rectangles and triangles turn into clean shapes.



Pen menu with Magic Highlighter selected

Magic Highlighter

Use it to point something out while you explain. Choose **Pen → Magic Highlighter** (the wand icon) and draw. The mark appears at once, stays for **about five seconds**, then **fades away by itself**.



A Magic Highlighter mark. It disappears on its own after a few seconds.

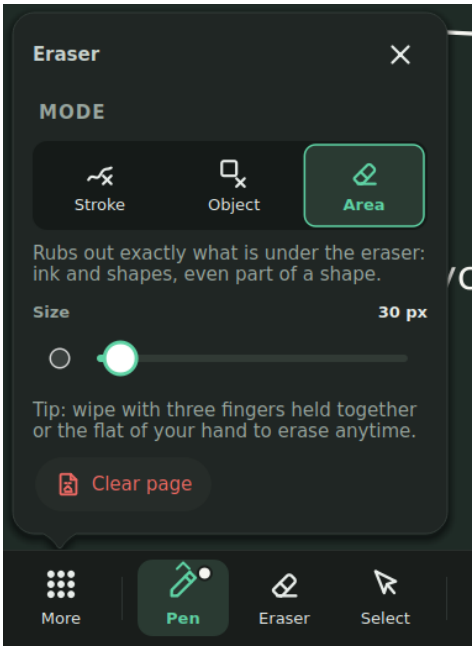
- Magic marks are **not part of the lesson**: they are not saved or exported, and Undo / Redo ignore them.
- It works with a finger, the pen and the mouse, and with several fingers at once.
- Choosing another style (Pen, Highlighter, …) switches back to normal ink. The normal **Highlighter** makes permanent, see-through marks.

5. Erasing

Tap **Eraser**, then tap it again for its menu. There are three modes:

- **Stroke** removes whole lines you touch.
- **Object** removes anything you touch (a line, shape, text, picture …).
- **Area** (normal) rubs out exactly what is under the eraser: ink and shapes.

Size sets the eraser size. **Clear page** removes everything on the page; it asks first and can be undone.



Eraser menu

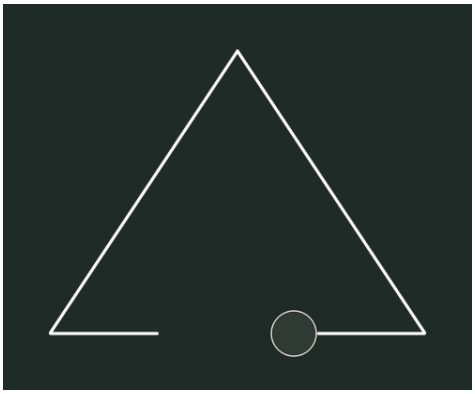
Erasing part of a shape

In **Area** mode you can rub out just part of a shape: one side of a triangle, half a side, or a gap in a circle. Rub over half a circle and a semicircle is left.

The part that is left becomes normal ink of the same colour and thickness. It looks the same but can no longer be resized as a shape. A fill (colour inside) is removed.

Arrows: the line can be cut anywhere; the arrowhead stays until you rub over it.

Undo brings the whole original shape back.



Part of a triangle's base erased

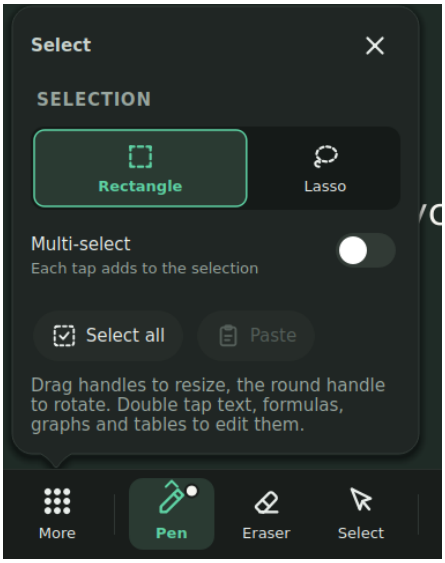
> **Tip:** For an exact semicircle or arc, use the **compass** (section 8) instead of erasing.

You can also wipe with **three fingers held together** at any time, without choosing the eraser (see section 3).

6. Selecting, moving and editing

Tap **Select**. Then:

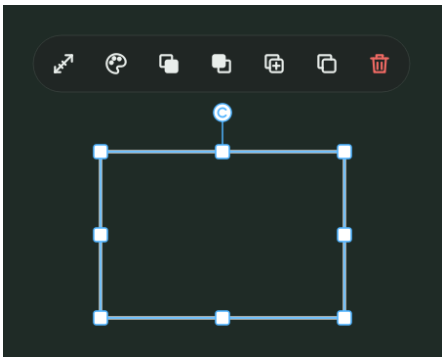
- **Tap** an object to select it (tapping inside an empty rectangle or circle works too).
- **Drag a rectangle** around objects, or switch to **Lasso** and draw around them.
- **Multi-select:** every tap adds to the selection.
- **Select all** and **Paste** are in the menu as well.



Select menu

With something selected:

- **Drag** it to move it.
- **Corner handles** resize it.
- The **round handle** rotates it (it snaps to 15°).
- Points of lines and polygons can be dragged one by one.
- The **floating bar** above it: edit, colour, bring to front / send to back, duplicate, copy, delete.
- **Double-tap** text, formulas, graphs or tables to edit them.

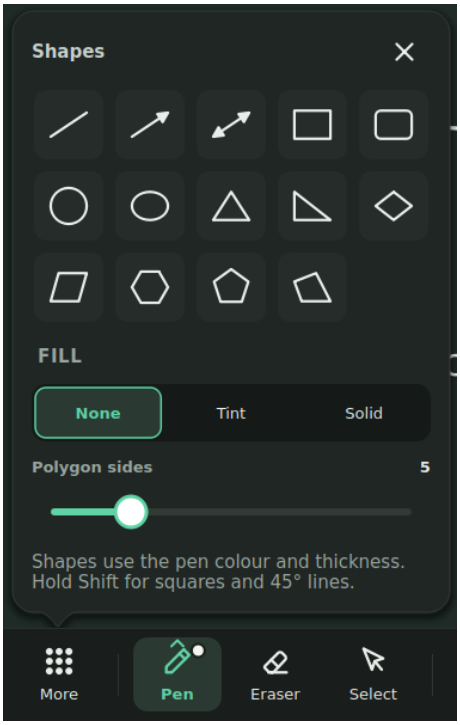


A selected rectangle with its handles and floating bar

7. Shapes, text, images and tables

More → Shapes. Tap a shape, then drag on the board (or just tap for a default size).

- Line, arrow, double arrow, rectangle, rounded rectangle, circle, ellipse, triangle, right triangle, diamond, parallelogram, hexagon, regular polygon and free polygon.
- **Fill:** None, Tint (see-through) or Solid. **Polygon sides** sets the regular polygon.
- Hold **Shift** (keyboard) for squares, circles and 45° lines.
- **Free polygon:** tap its corners, then tap the first corner again (or double-tap, or press Enter).
- Shapes use the pen colour and thickness.



Shapes menu

Text

More → Text, tap the board and type. Size, bold / italic / underline, alignment and colour apply to the text you are editing or to the selected text boxes. Double-tap a text box (with Select) to edit it again.

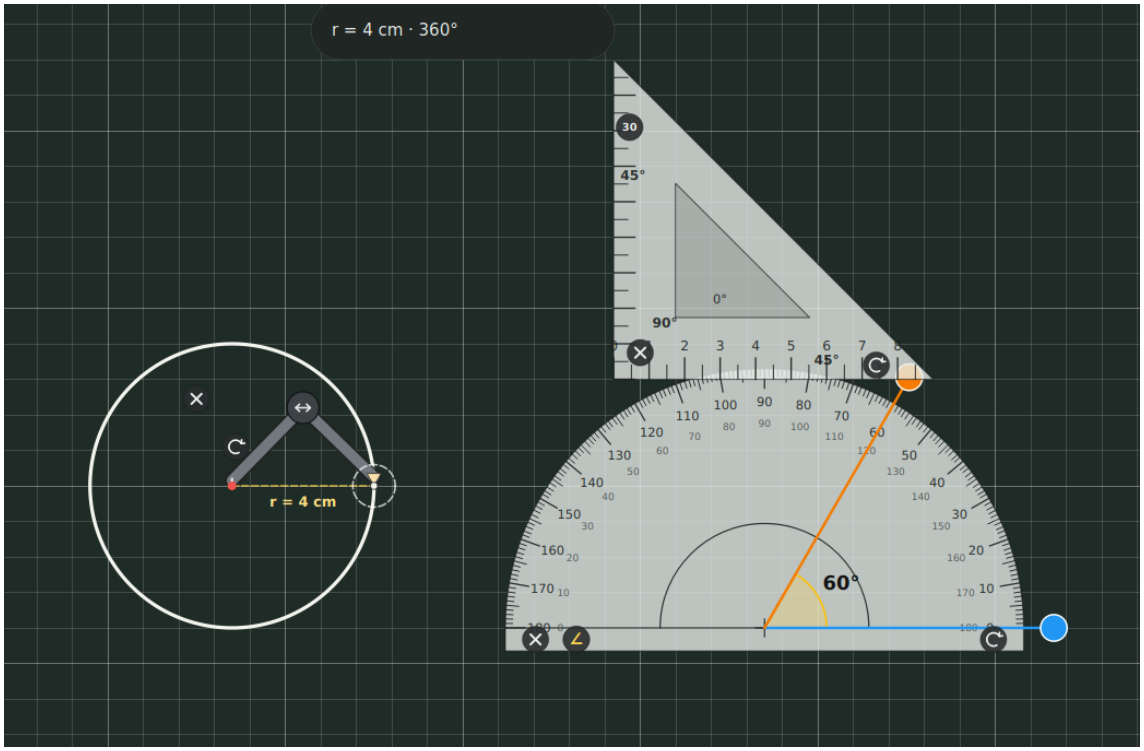
Images

More → Images or **Insert → Image**. You can also drag an image file onto the board, or paste one.

Tables

More → Table inserts a table. Double-tap a cell (with Select) to type in it.

8. Geometry: instruments, measurements and exact values



Compass, protractor and set square on a grid background

Instruments (ruler, protractor, set squares, compass)

- Open them from **More**. Drag the body to move an instrument; drag the round arrow to rotate it (snaps to 15°), or use two fingers to move and rotate at once. Tap ✕ to put it away.
- **Ruler and set square:** start drawing with the pen near an edge and the line follows the edge exactly. The length is shown while you draw. Drag the tab at the end of the ruler to change its length.
- **Protractor:** drag the blue and orange knobs to measure an angle; the ∠ button stamps the angle onto the page.
- **Compass:** drag the pencil to draw an arc or a full circle; drag the hinge to change the radius; drag the needle leg to move it.

Measurements and constructions

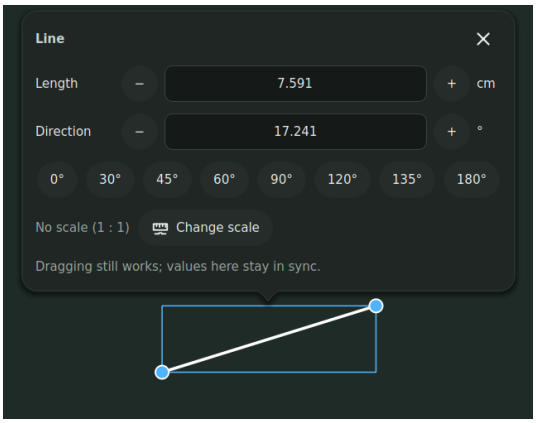
- **More → Measurements:** distance, angle, slope and area. The values update when you move the points.
- **More → Geometry:** points (named A, B, C … with coordinates), segments, lines, rays and vectors. They snap to existing points and, on grid backgrounds or inside graphs, to whole coordinates.
- **Coordinate plane background** (Templates) puts numbered axes on the page: 1 cm = one grid square.

Exact values

Select a line, vector, measurement, point or shape and tap the **precision** button in the floating bar (or double-tap it). Type or step exact values:

- length and direction of lines (0°, 30°, 45° … in one tap)
- rise, run, slope and slope angle
- a vector's size and direction
- a shape's width and height (area and perimeter are shown)

Dragging still works and the values follow.



Exact values of a line

Scale

In the property panel tap **Change scale** (or **More → Measurements → Scale**) and set, for example, **10 cm = 1 km**. Lengths, areas and vectors are then shown in real units (a 7 cm line reads 0.7 km), and you can type real values (1 km draws a 10 cm line).

9. Equations and the Magic Equation Maker

More → Equation. Type the formula or tap the symbol buttons; the preview updates as you type. Tap **Insert formula**.

Examples:

```
\frac{a}{b}   x^{2}   x_{n}   \sqrt{x}
\sqrt[3]{x}
\int_{0}^{1} f(x)\,dx   \sum_{i=1}^{n} i
\lim_{x \to 0}   \vec{v}   \alpha   \pi
```

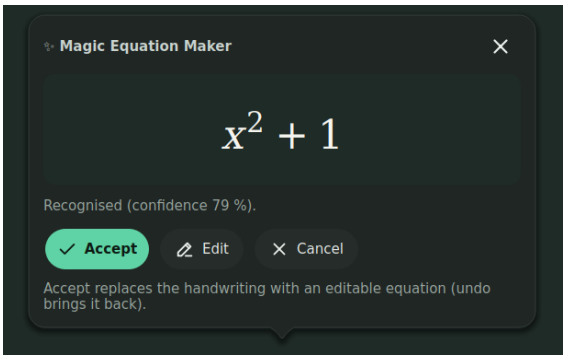
Matrices, cases and Greek letters work too. Double-tap a formula (with Select) to edit it.



The Equation menu

Magic Equation Maker: turn handwriting into an equation

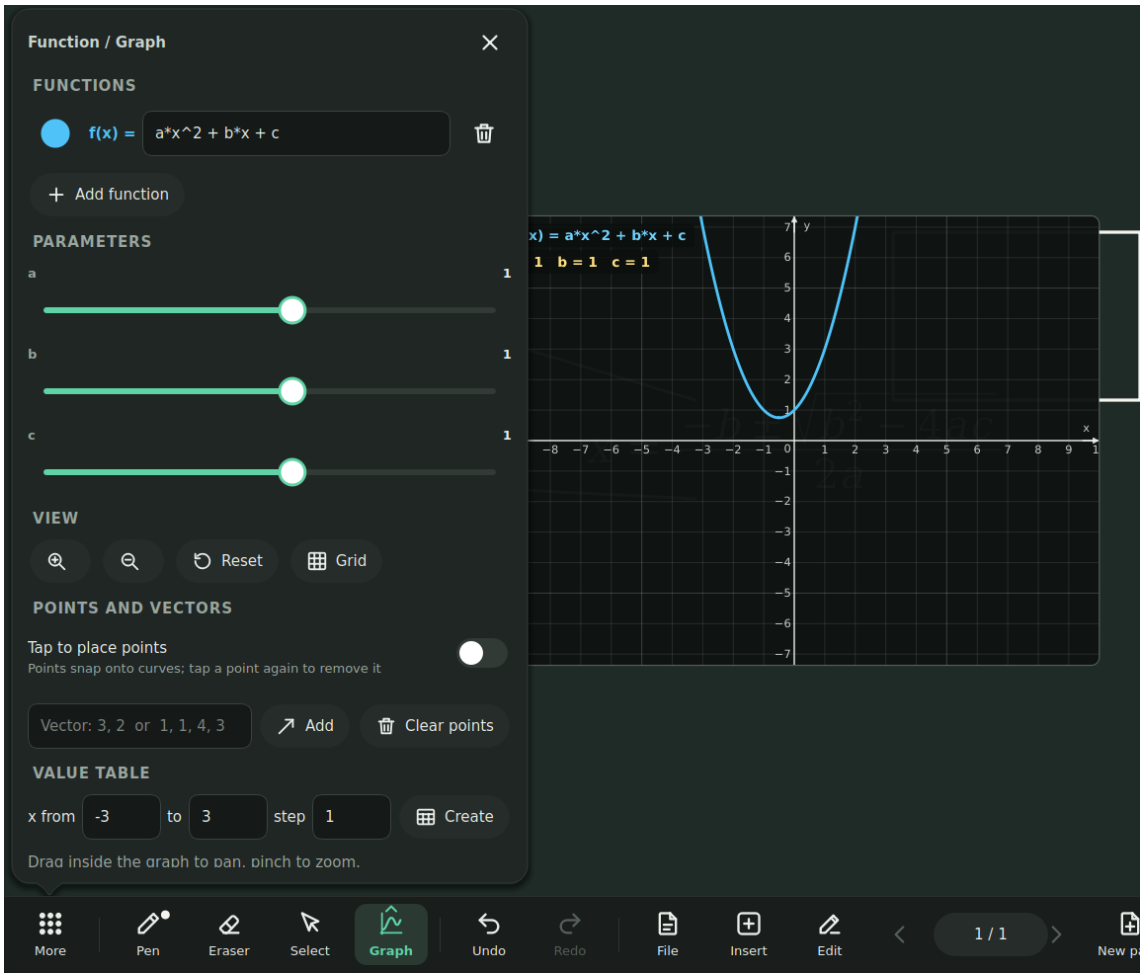
- Write an expression with the pen.
- Select the handwriting with **Select**, then tap the **magic wand** in the floating bar (or **Magic Equation Maker** in the Equation menu).
- Check the preview: **Accept** replaces the handwriting with an editable equation (one Undo brings the handwriting back), **Edit** lets you correct it first, **Cancel** changes nothing.



Preview before anything changes

If it can't read the handwriting it says so and offers **Try again**, **Edit manually** or **Cancel**; it never changes your writing on its own. It reads digits, x, y, a, b, c, n, + − = ( ), powers such as x² and simple fractions, all offline.

10. Graphs



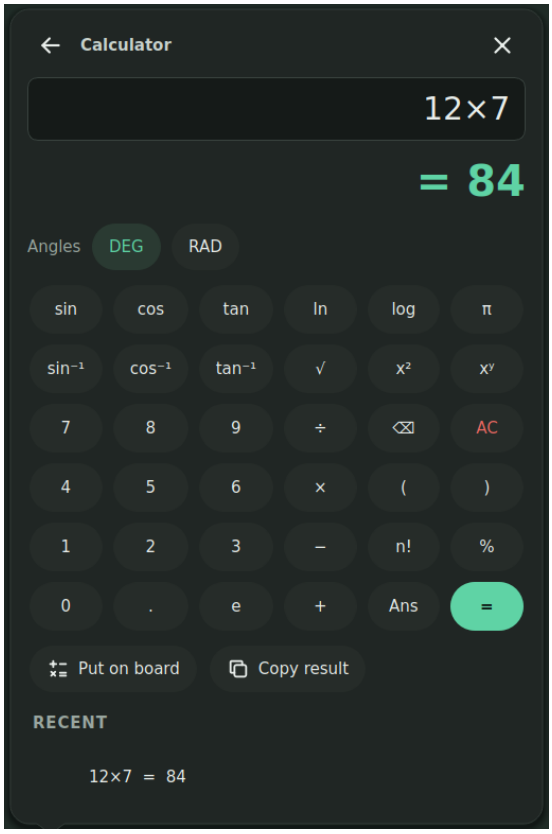
A graph with parameter sliders a, b and c

- **More → Function / Graph → Insert graph**.
- Type functions such as **2x + 1**, **a*sin(b x) + c**, **x² − 4**, **|x|**, **sqrt(x)** or **f(x) = e^x**. **Add function** plots several at once.
- Letters other than x become **parameters**, each with its own slider.
- Drag inside the graph to move it and pinch to zoom, or use the zoom buttons.
- **Tap to place points** adds points that snap onto the curves (tap a point again to remove it).
- Add **vectors** by typing their components, and create **value tables** for the visible functions.

11. Calculator

More → Calculator opens a scientific calculator with big touch keys. It works like a school calculator:

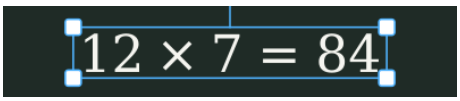
- Tap the keys, or type on a keyboard (Enter = equals, * and / work too).
- The answer appears **as you type**; **=** keeps it as **Ans** for the next calculation. After = an operator such as **×** continues with the answer, a number starts a new calculation.
- ⌫ deletes the last number or function, **AC** clears everything.
- **Recent** lists the last calculations: tap one to use its result again.



The Calculator (More → Calculator)

Keys	What they do	Example
+ − × ÷ ( )	Arithmetic; brackets are closed for you	(3 + 4) × 12 = 84
x² xʸ √	Square, any power, square root	√(2) = 1.41421356237
sin cos tan, sin⁻¹ cos⁻¹ tan⁻¹	Trigonometry in **DEG** (degrees, the default) or **RAD**	sin 30 = 0.5, tan⁻¹(1) = 45
ln log	Natural and base-10 logarithm	log(1000) = 3
π e	The constants	2π = 6.28318530718
n! %	Factorial and percent	5! = 120, 15% × 80 = 12
Ans	The last result	Ans ÷ 4 = 21

Put on board writes the calculation and its answer onto the page as an equation (for example 12 × 7 = 84). It is selected, so you can move or resize it straight away; Undo removes it. **Copy result** copies the number.



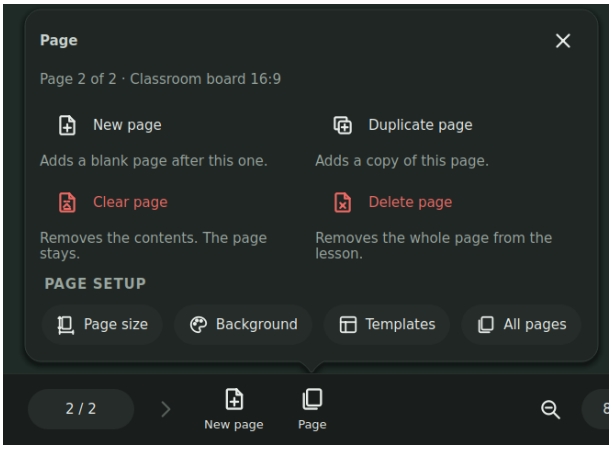
After Put on board

If something can't be calculated, the calculator says why in plain words, for example Not defined (√ of a negative number, 0 ÷ 0 or tan 90°) or Too large, or division by zero. The calculator keeps what you typed while it is closed.

12. Pages, backgrounds and templates

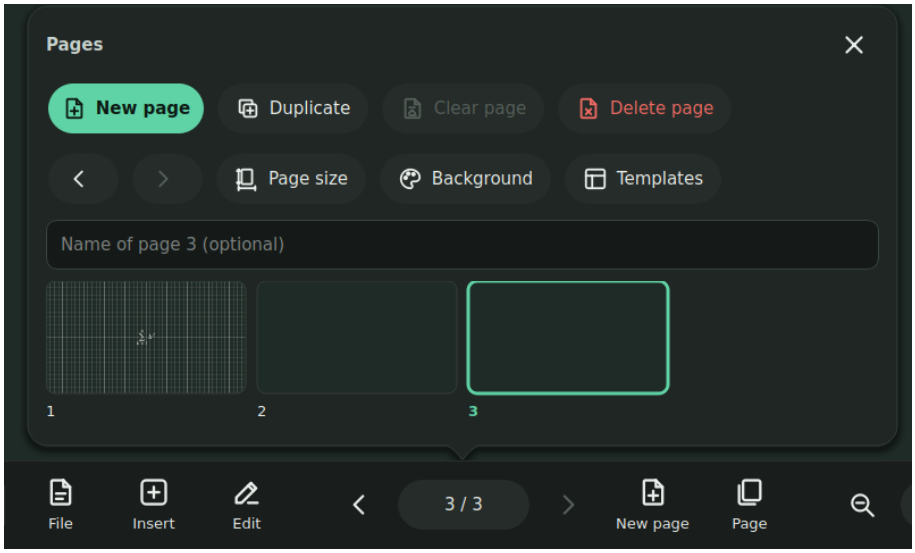
New page (ribbon) adds a blank page after the current one. The **Page** button opens the page menu:

- **New page**, **Duplicate page**
- **Clear page:** removes everything, keeps the page (asks first, can be undone)
- **Delete page:** removes the page itself (Undo in the message brings it back)
- **Page size**, **Background**, **Templates**



Page menu

- **Page size:** classroom board 16:9, 4:3, A4, A3 (landscape or portrait) or a custom size in cm, for this page, all pages or new pages. 1 cm on the page is always 1 cm in measurements.
- **Background:** white, black, light grey, blackboard green, more colours or any custom colour, for this page or all pages. Grid and line patterns are kept.
- **Templates** (More → Templates): blackboard, whiteboard, grid, dots, ruled, graph paper, music staves, coordinate plane, or your own image (Background from image, Save this background).



Page navigator: tap the page number in the ribbon

In the **page navigator** tap a thumbnail to open it, drag thumbnails to change the order (or use ◀ ▶), and duplicate, clear, delete or name a page.

13. Lessons: saving, importing and exporting

Save and open

- **File** (or More → File): new, open, save, save as and recent lessons. Lessons are **.classboard** files; double-clicking one opens it in ClassBoard.
- **Autosave** keeps a recovery copy while there are unsaved changes (the interval is in Settings). If ClassBoard or the computer stops unexpectedly, the next start offers to restore the lesson.

Import

What	How	Notes
Images	Insert → Image, More → Import → Images, drag onto the board, or paste	Import → Clipboard pastes what you copied.
Pages from another lesson	More → Import → Lesson pages	–
PDF documents	Insert → PDF, or drag the PDF onto the board	Every PDF page becomes a board page of the same size (e.g. A4) that you can write on.
PowerPoint (.ppt, .pptx)	Insert → PowerPoint, More → PPT / PPTX, or drag onto the board	Every slide becomes a page. Needs **Microsoft PowerPoint** or the free **LibreOffice** installed on the computer.

Export

More → Export: **PDF** (current page, whole lesson or pages such as 1-3, 5), **PowerPoint** (one slide per page) and **PNG** picture of the current page. Exports always contain the complete page, whatever the zoom.

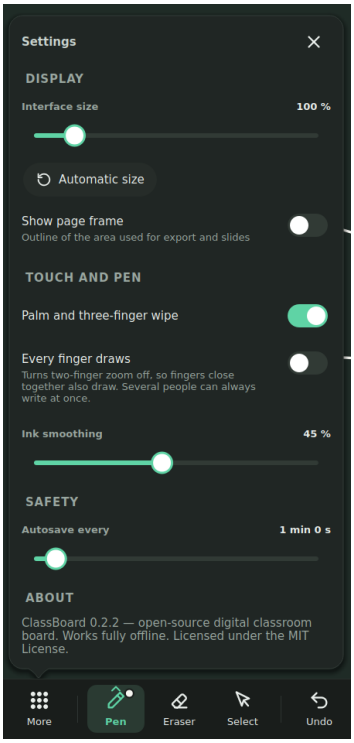
14. Zoom and view

- **−** and **+** step through the zoom levels (25 % to 250 %). Tap the **percentage** for the View menu: zoom presets, 100 %, fit page, fit width, full screen.
- The frame button fits the whole page; the last ribbon button switches full screen.
- Zoom only changes what you see: sizes, measurements and exports never change.

15. Settings

More → Settings:

- **Interface size** and **Automatic size**: make buttons and menus bigger or smaller.
- **Show page frame:** outline of the area used for export and slides.
- **Palm and three-finger wipe:** turn the wipe eraser on or off.
- **Every finger draws:** turns two-finger zoom off so fingers close together also draw.
- **Ink smoothing:** how much hand-drawn lines are smoothed.
- **Autosave every:** how often a recovery copy is saved.
- **About:** version, licence and third-party notices.



Settings

16. Keyboard shortcuts

Keys	Action
Ctrl+Z / Ctrl+Y (or Ctrl+Shift+Z)	Undo / redo
Ctrl+S / Ctrl+Shift+S / Ctrl+O / Ctrl+N	Save / save as / open / new lesson
Ctrl+C / Ctrl+X / Ctrl+V / Ctrl+D	Copy / cut / paste / duplicate
Ctrl+A, Delete	Select all, delete the selection
Arrow keys (Shift for bigger steps)	Nudge the selection
Page Up / Page Down, Ctrl+M	Previous / next page, new page
Ctrl+0, Ctrl++ / Ctrl+−	Fit page, zoom in / out
P, E, V, T	Pen, eraser, select, text
F11	Full screen
Esc	Close a menu, clear the selection or finish editing
Double-tap (with Select)	Edit text, formula, graph or table, or exact values of lines, measurements and shapes

17. Tips and troubleshooting

Problem	What to do
Windows says “Windows protected your PC”	Click **More info → Run anyway**. The installer is not signed; this is normal for this app.
Trying to zoom draws lines instead	Put the two fingers down closer together (one hand), or both at the same moment. Zoom does not start while someone else is writing.
Students zoom by accident	Turn on **Settings → Every finger draws**.
The wipe doesn't start	Hold three fingers **tightly together**. Check that **Palm and three-finger wipe** is on in Settings.
My hand draws while I use the pen	That shouldn't happen: touches are ignored while the pen is on or near the board. Lift the pen and your hand, then start again.
The eraser doesn't remove a shape	Use **Area** mode to rub out part of it, or **Object** mode to remove it with one tap.
I erased or deleted too much	Press **Undo**. It also brings back cleared and deleted pages.
PowerPoint import is greyed out	Install Microsoft PowerPoint or the free LibreOffice, then restart ClassBoard.
ClassBoard closed unexpectedly	Start it again: it offers to restore the lesson from autosave.

Good to know: Lessons are normal files in your Documents folder: copy them to a USB stick or cloud drive to use them on another board.

ClassBoard is free, open-source software under the MIT License. The Windows download also contains Qt (LGPL 3) and Poppler (GPL); their notices and licence texts are in the **doc** folder of the installation (THIRD_PARTY_NOTICES.md and licenses).